generators of random numbers; they are random number simulators. They are based on entirely classical deterministic mechanisms, using things like the digits of π to generate numbers. Nevertheless, they are good enough to fool you.

The game begins, and the computer continually updates the values of α_u and α_d. You wait as long as you want and then hit the M button. Then, with the aid of the random number generator, the game produces an outcome that is displayed on the screen. Based on this outcome, the computer updates the state by collapse. If the outcome is +1, the value of α_d is reset to zero, and the value of α_u is reset to unity. If the outcome is −1, the value of α_d is reset to unity, and the value of α_d is reset to zero. Then, the Schrödinger equation takes over until you hit M again.

Being a good experimenter, you do many trials and collect statistics, which you compare with quantum mechanical predictions. If everything works properly, you conclude that quantum mechanics is the correct description of whatever is taking place in the computer. Of course, the computer is still entirely classical, but it simulates a quantum spin without much difficulty.

Next, let's try the same thing with two computers, A and B, simulating two quantum spins. If the spins start in a product state and never interact, we can simply play the game on each of the two computers without any cross talk. But now, Alice, Bob, and Charlie return to help us out. Charlie, of course, wants to create an entangled pair. He begins by connecting the two computers with a cable to form a single computer, and we assume the cable can send instantaneous signals. In its memory, the combined computer now stores four complex numbers,

$$\alpha_{uu},\ \alpha_{ud},\ \alpha_{du},\ \alpha_{dd},$$

and it updates these numbers using the Schrödinger equation. Each computer screen shows an apparatus. Alice's screen shows $\mathcal{A}$ and Bob's screen shows $\mathcal{B}$. Each virtual apparatus can be independently oriented, and each can be